

1 An independence logic and its embedding

We use H_2Ol to give semantics and a soundness proof to a new logic, which we will refer to as IL , tailored to the propagation of *probabilistic independence* through an imperative program.

We introduce an assertion language $\{S \mid \phi\}$ that pairs an independence structure S with an ordinary first-order support predicate ϕ , equip it with a Hoare-style proof system over a family of imperative programs, and embed it into H_2Ol . Soundness of the object logic then follows by composing the embedding with the soundness of H_2Ol . We also show that the derived system helps us prove the variance-reduction guarantee of fedSGD.

1.1 The logic IL

Syntax. The logic will let us prove judgements over a family of programs written by the simple imperative programming language generated by this grammar:

$$c ::= x \leftarrow E \mid x \stackrel{\$}{\leftarrow} d \mid c; c \mid \text{for } k=a \text{ to } b \text{ do } c \mid \text{if } b \text{ then } c \text{ else } c.$$

As we mentioned, assertions in this logic will take the form $\{S \mid \phi\}$. Support formulas ϕ are just first-order formulas over expressions. Whereas, independence structures S are those generated by the following grammar:

$$S ::= \top \mid E * F \mid S|_b \mid S \wedge S.$$

where E, F range over expressions over program variables, and b over boolean expressions. The atom $E * F$ asserts independence of the two expressions; $S|_b$ is S *conditioned on* b , i.e. independence *conditional* on the guard b (it holds on the fibre where b is true); and $\wedge$ conjoins structures.

Semantics. IL is interpreted directly in H_2Ol : commands as terms of its programming language, and the satisfaction relations below as formulas of its ambient HOL . Commands are interpreted as follows:

$$\begin{aligned} \llbracket x \leftarrow E \rrbracket &= \lambda m. \text{return } m[x \mapsto \llbracket E \rrbracket_m], \\ \llbracket x \stackrel{\$}{\leftarrow} d \rrbracket &= \lambda m. v \stackrel{\$}{\leftarrow} \llbracket d \rrbracket_m; \text{return } m[x \mapsto v], \\ \llbracket c; c' \rrbracket &= \lambda m. m' \stackrel{\$}{\leftarrow} \llbracket c \rrbracket_m; \llbracket c' \rrbracket_{m'}, \\ \llbracket \text{if } b \text{ then } c \text{ else } d \rrbracket &= \lambda m. \text{if } \llbracket b \rrbracket_m \text{ then } \llbracket c \rrbracket_m \text{ else } \llbracket d \rrbracket_m, \\ \llbracket \text{for } k=a \text{ to } b \text{ do } c \rrbracket &= \lambda m. \text{iter}(b-a+1)(\lambda k \mu. m' \stackrel{\$}{\leftarrow} \mu; \llbracket c \rrbracket_{m'}) (\text{return } m). \end{aligned}$$

An independence structure is interpreted against a coupling μ . The relation $\mu \models S$, read “the coupling μ has independence structure S ”, is defined by induction on S :

$$\begin{aligned} \mu \models \top &\text{ always;} \\ \mu \models E * F &\iff |\mu| \cdot \text{bind } \mu \lambda \bar{m}. h(\llbracket E \rrbracket_{\bar{m}}, \llbracket F \rrbracket_{\bar{m}}) \\ &\quad = \text{bind } \mu \lambda \bar{m}_1. \text{bind } \mu \lambda \bar{m}_2. h(\llbracket E \rrbracket_{\bar{m}_1}, \llbracket F \rrbracket_{\bar{m}_2}) \quad \text{for all } h; \\ \mu \models S_1 \wedge S_2 &\iff \mu \models S_1 \text{ and } \mu \models S_2; \\ \mu \models S|_b &\iff \mu|_b \models S. \end{aligned}$$

Here $|\mu|$ is the total mass of μ and, for a boolean guard b , $\mu|_b$ is the subdistribution of μ supported on $\{\bar{m} \mid \llbracket b \rrbracket_{\bar{m}}\}$. The clause for $E * F$ is the independence of E and F under μ in scale-invariant form: the factor $|\mu|$ keeps it well defined on the subdistributions $\mu|_b$ produced by conditioning (for

couplings of mass 1 it drops out, leaving the usual factorisation). The clause for $S|_b$ thus reads independence *conditional* on b ; when $\mathbb{P}_\mu[b] = 0$ it holds vacuously, as $\mu|_b = 0$.

An assertion is satisfied by a coupling μ , written $\mu \models \{S \mid \phi\}$, if $\mu \models S$ and the support formula holds at every point of the support, $\forall \bar{m} \in \text{supp}(\mu). \bar{m} \models \phi$.

Definition 1 (Semantic validity, $\models$). A family of marginals $(\mu_i)_{i \in I}$ satisfies an assertion, written $(\mu_i)_{i \in I} \models \{S \mid \phi\}$, if some coupling of the family does:

$$(\mu_i)_{i \in I} \models \{S \mid \phi\} \triangleq \exists \mu. \mu \text{ couples } (\mu_i)_i \wedge \mu \models \{S \mid \phi\}.$$

Then

$$\models \{S \mid \phi\} [i: c_i]_{i \in I} \{S' \mid \phi'\} \triangleq \forall (\mu_i)_{i \in I}. (\mu_i)_i \models \{S \mid \phi\} \Rightarrow (m \stackrel{\$}{\leftarrow} \mu_i; \llbracket c_i \rrbracket m)_i \models \{S' \mid \phi'\}.$$

Proof system.

$$\begin{array}{c} \frac{}{\vdash \{S[x/\bar{E}] \mid \phi[x/\bar{E}]\} [i: x \leftarrow E_i]_{i \in I} \{S \mid \phi\}}^{\text{[ASSIGN]}} \\[10pt] \frac{\mu \triangleright d_1 \& \dots \& d_n \quad \forall \nu \models S. \text{bind } \nu \lambda \bar{m}. \text{bind } (\mu \bar{m}) \lambda \bar{v}. \text{unit}(\bar{m}[x/\bar{v}]) \models S'}{\vdash \{S \mid \forall \bar{v} \in \text{supp}(\mu). \phi[x/\bar{v}]\} [i: x \stackrel{\$}{\leftarrow} d_i]_{i \in I} \{S' \mid \phi\}}^{\text{[SAMPLE]}} \\[10pt] \frac{J \subseteq I \quad \mu \triangleright d_1 \& \dots \& d_n \quad \forall i \neq j \in J. \exists e_i, e_j. \forall \bar{m}. \pi_{\{i,j\}}(\mu \bar{m}) = e_i \otimes e_j}{\vdash \{\bigwedge_{i \neq j \in J} A_i * A_j \mid \forall \bar{v} \in \text{supp}(\mu). \phi[x/\bar{v}]\} [i \in I: x \stackrel{\$}{\leftarrow} d_i] \{\bigwedge_{i \neq j \in J} (A_i, x_i) * (A_j, x_j) \mid \phi\}}^{\text{[SAMPLE}_2\text{]}} \\[10pt] \frac{S' \vdash S \quad \phi' \Rightarrow \phi \quad \vdash \{S \mid \phi\} [i: c_i]_{i \in I} \{T \mid \psi\} \quad T \vdash S'' \quad \psi \Rightarrow \phi''}{\vdash \{S' \mid \phi'\} [i: c_i]_{i \in I} \{S'' \mid \phi''\}}^{\text{[CONSEQ]}} \\[10pt] \frac{\vdash \{S \mid \phi\} [i: c_i]_{i \in I} \{S' \mid \phi'\} \quad \vdash \{S' \mid \phi'\} [i: c'_i]_{i \in I} \{S'' \mid \phi''\}}{\vdash \{S \mid \phi\} [i: c_i; c'_i]_{i \in I} \{S'' \mid \phi''\}}^{\text{[SEQ]}} \\[10pt] \frac{\forall k \in \{a \dots b\}. \vdash \{\mathcal{J}(k)\} [i: c_i]_{i \in I} \{\mathcal{J}(k+1)\}}{\vdash \{\mathcal{J}(a)\} [i: \text{for } k=a \text{ to } b \text{ do } c_i]_{i \in I} \{\mathcal{J}(b+1)\}}^{\text{[FOR]}} \\[10pt] \frac{\phi \vdash b_1 = \dots = b_n =: b \quad \vdash \{S \mid \phi \wedge b\} [i: c_i]_{i \in I} \{T \mid \phi^c\} \quad \vdash_{\text{HOL}} \phi^c \Rightarrow p \quad \vdash \{S \mid \phi \wedge \neg b\} [i: d_i]_{i \in I} \{U \mid \phi^d\} \quad \vdash_{\text{HOL}} \phi^d \Rightarrow \neg p}{\vdash \{S|_b \wedge S|_{\neg b} \mid \phi\} [i: \text{if } b_i \text{ then } c_i \text{ else } d_i]_{i \in I} \{T|_p \wedge U|_{\neg p} \mid (p \Rightarrow \phi^c) \wedge (\neg p \Rightarrow \phi^d)\}}^{\text{[IF]}} \end{array}$$

In rules [SAMPLE] and [SAMPLE₂], μ is a store-indexed coupling of the samplers, i.e. a function from tuples $\bar{m} = (m_i)_{i \in I}$ of the stores on which the $\llbracket c_i \rrbracket$ act to distributions over tuples of sampled values, and

$$\mu \triangleright d_1 \& \dots \& d_n \triangleq \forall \bar{m}. \forall i \in I. \pi_i(\mu \bar{m}) = \llbracket d_i \rrbracket_{m_i}$$

with π_i the i -th marginal. In support formulas, μ stands for $\mu \bar{m}$ at the current store tuple $\bar{m}$. In rule [SAMPLE₂], $e \otimes e' \triangleq \text{bind } e \lambda u. \text{bind } e' \lambda w. \text{unit}(u, w)$ denotes the product of two distributions.

Assertion logic. We present some structural facts about S which may be useful to use with the program logic through [CONSEQ].

$$\begin{array}{ll}
[\text{AFFINE}] \quad S \wedge (E * F) \vdash S, & [\text{MAP}] \quad \overline{E} * \overline{F} \vdash f(\overline{E}) * g(\overline{F}), \\
[\text{COND-}\wedge] \quad (S_1 \wedge S_2)|_b \dashv\vdash S_1|_b \wedge S_2|_b, & [\text{COND-COND}] \quad (S|_{b'})|_b \dashv\vdash S|_{b \wedge b'}, \\
[\text{CONST}] \quad \top \vdash E * c \quad (c \text{ constant}). &
\end{array}$$

1.2 Embedding into $\mathbf{H_2O\ell}$

Writing $\llbracket \{S \mid \phi\} \rrbracket(\bar{\mu})$ for the HOL formula $(\mu_i)_i \models \{S \mid \phi\}$ of §1, we interpret an IL triple as follows:

$$\llbracket \vdash \{S \mid \phi\} [i: c_i]_{i \in I} \{S' \mid \phi'\} \rrbracket \triangleq (\mu_i : \mathbf{D} \sigma)_i \mid \llbracket \{S \mid \phi\} \rrbracket(\mu_i)_i \vdash [i: m \stackrel{\$}{\leftarrow} \mu_i; \llbracket c_i \rrbracket m]_{i \in I} \{\llbracket \{S' \mid \phi'\} \rrbracket\}.$$

Theorem 1 (Embedding of IL in $\mathbf{H_2O\ell}$). *If $\vdash \{S \mid \phi\} [i: c_i]_{i \in I} \{S' \mid \phi'\}$, then $\llbracket \vdash \{S \mid \phi\} [i: c_i]_{i \in I} \{S' \mid \phi'\} \rrbracket$ is derivable in $\mathbf{H_2O\ell}$.*

The proof is presented in Appendix A.

Corollary 1 (Soundness of IL). *If $\vdash \{S \mid \phi\} [i: c_i]_{i \in I} \{S' \mid \phi'\}$ then $\models \{S \mid \phi\} [i: c_i]_{i \in I} \{S' \mid \phi'\}$.*

1.3 Application: variance reduction in federated SGD

Federated SGD has n parties collaboratively train a shared model: at each round, every party computes a gradient on a minibatch drawn from its own local dataset D_i , and a central server aggregates these per-party gradients with fixed weights $w_1, \dots, w_n$. When the parties' datasets are independent, the aggregate enjoys a variance reduction over any single solo run of SGD. Let $W = \sum_j w_j$, let the affine gradient be $\nabla f(\theta, x) = \theta - x$ (square loss), and write the batch update $U_B(\theta) \triangleq \theta - \eta \frac{1}{m} \sum_{x \in B} \nabla f(\theta, x)$. The index set is $I = \{1, \dots, n, n+1\}$: for $i \leq n$ the i -th program runs **solo** on the local dataset D_i ; the $(n+1)$ -th program runs **fed** on $\overline{D} = (D_1, \dots, D_n)$. There are R rounds.

$$\begin{array}{ll}
\text{solo}(\theta_{\text{init}}, D): & \text{fed}(\theta_{\text{init}}, \overline{D}, \overline{w}): \\
\theta \leftarrow \theta_{\text{init}}; & \theta \leftarrow \theta_{\text{init}}; \\
\text{for } k=1 \text{ to } R \text{ do} & \text{for } k=1 \text{ to } R \text{ do} \\
\quad B \stackrel{\$}{\leftarrow} \text{minibatch } m \ D; & \quad B \stackrel{\$}{\leftarrow} \text{minibatchN } m \ \overline{D} \quad (B = (B_1, \dots, B_n)); \\
\quad \theta \leftarrow \theta - \eta \frac{1}{m} \sum_{x \in B} \nabla f(\theta, x) & \quad \theta \leftarrow \theta - \eta \sum_{k \leq n} \frac{w_k}{W} \frac{1}{m} \sum_{x \in B_k} \nabla f(\theta, x)
\end{array}$$

We write θ_i ($i \leq n$), θ_{n+1} for the models, B_i for the batch of program $i \leq n$, and $(B_1^f, \dots, B_n^f)$ for the batch tuple of program $n+1$. The assertion we want to reach is

$$\mathcal{I} \triangleq \left\{ \underbrace{\bigwedge_{i \neq j \leq n} \theta_i * \theta_j}_S \mid \underbrace{\sum_{i \leq n} \frac{w_i}{W} \theta_i = \theta_{n+1}}_{\phi} \right\} :$$

the local models are pairwise independent, and the federated model is their weighted mean. The body of each program is $c_i^{\text{sm}}; c_i^{\text{up}}$, where the sampling command is $c_i^{\text{sm}} = (B \stackrel{\$}{\leftarrow} d_i)$ with $d_i = \text{minibatch } m \ D_i$ for $i \leq n$ and $d_{n+1} = \text{minibatchN } m \ \overline{D}$, and the update command is $c_i^{\text{up}} = (\theta \leftarrow U_B(\theta))$, that is $\theta \leftarrow U_{B_i}(\theta_i)$ for $i \leq n$ and $\theta \leftarrow U^{\text{fed}}(\theta_{n+1})$ for the federated program, with $U^{\text{fed}}(\theta_{n+1}) \triangleq \theta_{n+1} - \eta \sum_{k \leq n} \frac{w_k}{W} \frac{1}{m} \sum_{x \in B_k^f} \nabla f(\theta_{n+1}, x)$.

We show $\vdash \{ \top \mid \top \} \bar{c} \mathcal{I}$, where $\bar{c} = [i: c_i]_{i \in I}$ is the family with $c_i = \text{solo}(\theta_{\text{init}}, D_i)$ for $i \leq n$ and $c_{n+1} = \text{fed}(\theta_{\text{init}}, \bar{D}, \bar{w})$. Applying [SEQ] to split the initialisation from the loop, the initialisation establishes

$$\vdash \{ \top \mid \top \} [i: \theta \leftarrow \theta_{\text{init}}]_{i \in I} \mathcal{I},$$

by [ASSIGN] with [CONSEQ]: the substituted precondition $\mathcal{I}[\theta/\theta_{\text{init}}]$ has structure $\bigwedge_{i \neq j \leq n} \theta_{\text{init}} * \theta_{\text{init}}$, entailed by $\top$ through [CONST], and support $\sum_{i \leq n} \frac{w_i}{W} \theta_{\text{init}} = \theta_{\text{init}}$, valid since $\sum_i w_i = W$. Then [FOR] with the round-independent invariant $\mathcal{J}(k) \equiv \mathcal{I}$ reduces the goal to the single-iteration obligation

$$\vdash \mathcal{I} [i: c_i^{\text{sm}} ; c_i^{\text{upd}}]_{i \in I} \mathcal{I}. \quad (*)$$

For the sampling command,

$$\vdash \mathcal{I} [i: c_i^{\text{sm}}]_{i \in I} \{ S_1 \mid \phi \wedge \bigwedge_{k \leq n} B_k^{\text{f}} = B_k \}, \quad (1)$$

with $S_1 = \bigwedge_{i \neq j \leq n} (\theta_i, B_i) * \{\theta_j, B_j\}$, by [SAMPLE₂] with $J = \{1, \dots, n\}$ and $A_i = \theta_i$, instantiated at the coupling μ that draws $B_1, \dots, B_n$ from $\bigotimes_{k \leq n} \text{minibatch}(D_k)$ and returns $(B_1, \dots, B_n)$ as program $n+1$'s batch tuple. This μ meets the premises of [SAMPLE₂] and on its entire support satisfies $B_k^{\text{f}} = B_k$ for $k \leq n$. Appending each fresh B_i to its group A_i produces the structure S_1 ; and, taking postcondition support formula $\phi \wedge \bigwedge_{k \leq n} B_k^{\text{f}} = B_k$, the precondition required by [SAMPLE₂] is $\forall \bar{v} \in \text{supp} \mu. (\phi \wedge \bigwedge_k v_k^{\text{f}} = v_k)$, where $\bar{v}$ ranges over the jointly sampled batch tuples. Distributing the quantifier over the conjunction, this is $(\forall \bar{v} \in \text{supp} \mu. \phi) \wedge (\forall \bar{v} \in \text{supp} \mu. \bigwedge_k v_k^{\text{f}} = v_k)$: the first conjunct is ϕ , since ϕ contains no batch variable and so does not depend on $\bar{v}$; the second is valid, since every $\bar{v} \in \text{supp} \mu$ satisfies $v_k^{\text{f}} = v_k$. The precondition is thus ϕ , so we get the invariant $\mathcal{I}$.

For the update command,

$$\vdash \{ S_2 \mid \psi \} [i: c_i^{\text{upd}}]_{i \in I} \mathcal{I}, \quad (2)$$

where $S_2 = \bigwedge_{i \neq j \leq n} U_{B_i}(\theta_i) * U_{B_j}(\theta_j)$ and $\psi = (\sum_{i \leq n} \frac{w_i}{W} U_{B_i}(\theta_i) = U^{\text{fed}}(\theta_{n+1}))$, directly by [ASSIGN], since $\{ S_2 \mid \psi \}$ is $\mathcal{I}$ with each θ_i replaced by the update it receives.

Lemma 1 (Linearity of the aggregate). *If $\sum_{i \leq n} \frac{w_i}{W} \theta_i = \theta_{n+1}$ and $B_k^{\text{f}} = B_k$ for all $k \leq n$, then $\sum_{i \leq n} \frac{w_i}{W} U_{B_i}(\theta_i) = U^{\text{fed}}(\theta_{n+1})$.*

Proof. With $\nabla f(\theta, x) = \theta - x$ we have $U_B(\theta) = (1 - \eta)\theta + \eta \bar{B}$, $\bar{B} = \frac{1}{m} \sum_{x \in B} x$. Then $\sum_i \frac{w_i}{W} U_{B_i}(\theta_i) = (1 - \eta)\theta_{n+1} + \eta \sum_i \frac{w_i}{W} \bar{B}_i$ using $\sum_i \frac{w_i}{W} \theta_i = \theta_{n+1}$, while $U^{\text{fed}}(\theta_{n+1}) = (1 - \eta)\theta_{n+1} + \eta \sum_k \frac{w_k}{W} \bar{B}_k^{\text{f}}$ (as $\sum_k \frac{w_k}{W} = 1$); the two agree since $B_k^{\text{f}} = B_k$. $\square$

Composing (1) and (2) closes the argument. The postcondition of (1) entails the precondition of (2): its structure S_1 gives S_2 by [MAP], because $U_{B_i}(\theta_i)$ is a function of the pair (θ_i, B_i) ; and its support formula $\phi \wedge \bigwedge_{k \leq n} B_k^{\text{f}} = B_k$ gives ψ by Lemma 1. Hence [CONSEQ] turns (1) into $\vdash \mathcal{I} [i: c_i^{\text{sm}}]_{i \in I} \{ S_2 \mid \psi \}$, and [SEQ] with (2) yields the single-iteration obligation (*), establishing $\vdash \mathcal{I} \bar{c} \mathcal{I}$.

With pairwise independence S and the weighted-mean relation ϕ in force at the end, the variance reduction $\text{Var}(\theta_{n+1}) = \sum_{i \leq n} \frac{w_i}{W}^2 \text{Var}(\theta_i)$ follows as an HOL consequence.

A Soundness of the embedding

We prove Theorem 1 by induction on the derivation of the IL triple, showing that each rule translates to a H₂O ℓ derivation of the interpreted judgment. A substitution lemma supports the assignment case.

Lemma 2 (Substitution). *For every assertion A ,*

$$(\mu_i)_i \models A[x/\overline{E}] \Leftrightarrow (m \stackrel{\$}{\leftarrow} \mu_i; \mathbf{return} \ m[x \mapsto \llbracket E_i \rrbracket_m])_i \models A.$$

Proof. By induction on A : expression substitution commutes with the pushforward along the deterministic assignment $\lambda m. \mathbf{return} \ m[x \mapsto \llbracket E_i \rrbracket_m]$; the atoms $E * F$, the conditioning $S|_b$ (substituting into the guard as well), and the support formula are each stable under this pushforward. $\square$

Proof of Theorem 1. By induction on the derivation. The interpreted judgment has the input marginals (μ_i) as free variables, the interpreted precondition $\llbracket \{S \mid \phi\} \rrbracket (\mu_i)_i$ as its logical context Φ , and the output family $(m \stackrel{\$}{\leftarrow} \mu_i; \llbracket c_i \rrbracket \ m)$ as the output distributions; under Φ we derive the postcondition $\llbracket \{S' \mid \phi'\} \rrbracket$ of that family, discharging each case by the corresponding $\mathbf{H}_2\mathbf{O}\ell$ rule applied to the monadic terms $m \stackrel{\$}{\leftarrow} \mu_i; \llbracket c_i \rrbracket \ m$.

[ASSIGN] $\llbracket c_i \rrbracket = \lambda m. \mathbf{return} \ m[x \mapsto \llbracket E_i \rrbracket_m]$ is a deterministic **return**, so by **MUNIT** the output is the pushforward $m \stackrel{\$}{\leftarrow} \mu_i; \llbracket c_i \rrbracket \ m$. By Lemma 2, $(\mu_i)_i \models \{S[x/\overline{E}] \mid \phi[x/\overline{E}]\} \Leftrightarrow (m \stackrel{\$}{\leftarrow} \mu_i; \llbracket c_i \rrbracket \ m)_i \models \{S \mid \phi\}$, which is the required implication, discharged by **CONSEQ**.

[SAMPLE] $\llbracket c_i \rrbracket = \lambda m. v \stackrel{\$}{\leftarrow} \llbracket d_i \rrbracket \ m; \mathbf{return} \ m[x \mapsto v]$ is a monadic bind, discharged by **MBIND** with the coupling μ of the rule instance as witness, whose marginals are the samplers by the first premise. **MBIND** establishes the structure S' from its premise on the composite coupling, which is the second premise of **[SAMPLE]**,

$$\forall \nu \models S. \mathbf{bind} \ \nu \ \lambda \overline{m}. \mathbf{bind} \ (\mu \ \overline{m}) \ \lambda \overline{v}. \mathbf{unit}(\overline{m}[x/\overline{v}]) \models S';$$

the support ϕ follows from the precondition $\forall \overline{v} \in \mathbf{supp} \mu. \phi[x/\overline{v}]$, which holds at every sampled value and hence at every point $\overline{m}[x/\overline{v}]$ of the output.

[SAMPLE₂] This specialises **[SAMPLE]** to $S = \bigwedge_{i \neq j \in J} A_i * A_j$ and $S' = \bigwedge_{i \neq j \in J} (A_i, x_i) * (A_j, x_j)$: the sampling is the same monadic bind, discharged by **MBIND**, so it remains to check **[SAMPLE]**'s second premise: $\rho \models S'$ for every $\nu \models S$, where $\rho = \mathbf{bind} \ \nu \ \lambda \overline{m}. \mathbf{bind} \ (\mu \ \overline{m}) \ \lambda \overline{v}. \mathbf{unit}(\overline{m}[x/\overline{v}])$. Fix $i \neq j \in J$. Since $\pi_{\{i,j\}}(\mu \ \overline{m}) = e_i \otimes e_j$ for every $\overline{m}$, the fresh x_i, x_j are drawn from a fixed product, so under ρ they are independent of the pre-state pair (A_i, A_j) and of each other; with $A_i * A_j$ from S , the bundles (A_i, x_i) and (A_j, x_j) are independent, i.e. $(A_i, x_i) * (A_j, x_j)$. The relations of the programs in $I \setminus J$ travel in ϕ through $\forall \overline{v} \in \mathbf{supp} \mu.$, as in **[SAMPLE]**.

[SEQ] $\llbracket c_i ; c'_i \rrbracket = \lambda m. m' \stackrel{\$}{\leftarrow} \llbracket c_i \rrbracket \ m; \llbracket c'_i \rrbracket \ m'$ is Kleisli composition; the case follows from the two induction hypotheses and **MBIND**.

[FOR] $\llbracket \mathbf{for} \dots \rrbracket = \lambda m. \mathbf{iter} \ (b-a+1) \ (\lambda k \mu. m' \stackrel{\$}{\leftarrow} \mu; \llbracket c_i \rrbracket \ m') \ (\mathbf{return} \ m)$; by **ITER** of $\mathbf{H}_2\mathbf{O}\ell$ with invariant $\llbracket \mathcal{J}(k) \rrbracket$, whose step premise is the induction hypothesis $\{i: c_i\} \{ \mathcal{J}(k+1) \}$.

[CONSEQ] $\mathbf{H}_2\mathbf{O}\ell$'s **CONSEQ**. The object side-conditions translate to **HOL** entailments under $\llbracket \cdot \rrbracket$: the structure entailments $S' \vdash S$ and $T \vdash S''$ give $\llbracket S' \rrbracket \Rightarrow \llbracket S \rrbracket$ and $\llbracket T \rrbracket \Rightarrow \llbracket S'' \rrbracket$ by soundness of the assertion logic (each rule **[AFFINE]**, **[MAP]**, **[COND- $\wedge$]**, **[COND-COND]**, **[CONST]** is **HOL**-valid), while $\phi' \Rightarrow \phi$ and $\psi \Rightarrow \phi''$ are **HOL** implications on the support. $\mathbf{H}_2\mathbf{O}\ell$'s **CONSEQ** then strengthens the precondition and weakens the postcondition.

[IF] Under synchronised guards $(\phi \vdash b_1 = \dots = b_n =: b)$ the family denotes the branch mixture $\rho = \rho_c \uplus \rho_d$, with $\rho_c = \llbracket [i: c_i] \rrbracket (\mu|_b)$ and $\rho_d = \llbracket [i: d_i] \rrbracket (\mu|_{\neg b})$. Since **if** b **then** c **else** $d = \mathbf{case} \ b \ \mathbf{of} \ \mathbf{tt} \rightarrow c \mid \mathbf{ff} \rightarrow d$ (on **Bool** = $1 + 1$), the family is discharged by $\mathbf{H}_2\mathbf{O}\ell$'s lock-step rule

CASE, with guard relation $\psi(\bar{v}) \triangleq (v_1 = \dots = v_n)$, which holds by the synchronisation premise and makes every non-constant branch vector contradictory; the premises of the two constant vectors are the induction hypotheses. It remains to check the postcondition $\varphi \triangleq \llbracket \{T|_p \wedge U|_{\neg p} \mid (p \Rightarrow \phi^c) \wedge (\neg p \Rightarrow \phi^d)\} \rrbracket$ of the output ρ . By the induction hypothesis $\rho_c \models \{T \mid \phi^c\}$, so ϕ^c holds on $\text{supp}\rho_c$; with the side condition $\vdash_{\text{HOL}} \phi^c \Rightarrow p$ this gives $\text{supp}\rho_c \subseteq \{\bar{m} \mid \llbracket p \rrbracket_{\bar{m}}\}$, and symmetrically $\text{supp}\rho_d \subseteq \{\bar{m} \mid \neg \llbracket p \rrbracket_{\bar{m}}\}$. Restriction is pointwise, so it distributes over the mixture and each summand is recovered exactly:

$$\rho|_p = \rho_c|_p \uplus \rho_d|_p = \rho_c \uplus 0 = \rho_c, \quad \rho|_{\neg p} = \rho_d.$$

Hence $\rho \models T|_p$ and $\rho \models U|_{\neg p}$ are the two induction hypotheses, and the support conjunct $(p \Rightarrow \phi^c) \wedge (\neg p \Rightarrow \phi^d)$ holds at every point of $\text{supp}\rho = \text{supp}\rho_c \cup \text{supp}\rho_d$, each implication being witnessed on one side and vacuous on the other. CASE concludes φ of the output.

This exhausts the rules, proving the theorem. Corollary 1 then follows by Theorem 4.2: for the interpreted judgment its conclusion $\Gamma \models \bigwedge \Phi \Rightarrow \phi \bar{s}$ unfolds to

$$\forall(\mu_i)_i. (\mu_i)_i \models \{S \mid \phi\} \Rightarrow (m \stackrel{*}{\leftarrow} \mu_i; \llbracket c_i \rrbracket m)_i \models \{S' \mid \phi'\},$$

which is Definition 1. □